

## Day 8 — Tirthaji's Sūtra System in Context

**Module:**  $\times 11$  (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Time:** 45 min

### Learning objective

By the end of this lesson, students will name at least 5 of Tirthaji's 16 sūtras, explain in 2–3 sentences when and by whom *Vedic Mathematics* was published, and state honestly that the “Vedic” attribution is academically contested.

### Materials

- Printed handout: the list of 16 sūtras + 13 sub-sūtras (from Microsoft Word - sutras1.pdf).
- Optional: a paragraph excerpt from Tirthaji's preface (teacher's prepared synopsis).

### Lesson flow

#### Warm-up (3 min)

- Recall: name the sūtra behind the  $\times 11$  trick. (*Ekādhikena Pūrveṇa*.)
- Name the sub-sūtra behind  $\times 12$ – $\times 19$ . (*Anurūpyeṇa*.)

#### Core (25 min) — The 16 sūtras and the provenance question

- The 16 sūtras.** Hand out the list. Walk through abbreviated:

| #  | Sanskrit                               | English gloss                                |
|----|----------------------------------------|----------------------------------------------|
| 1  | <i>Ekādhikena Pūrveṇa</i>              | “By one more than the previous”              |
| 2  | <i>Nikhilam Navataścaramaṁ Daśataḥ</i> | “All from 9 and the last from 10”            |
| 3  | <i>Ūrdhva-tiryagbhyāṁ</i>              | “Vertically and crosswise”                   |
| 4  | <i>Parāvartya Yojayet</i>              | “Transpose and apply”                        |
| 5  | <i>Śūnyaṁ Sāmyasamuccaye</i>           | “When the sum is the same, that sum is zero” |
| 6  | <i>(Ānurūpye) Śūnyam Anyat</i>         | “If one is in ratio, the other is zero”      |
| 7  | <i>Sankalana Vyavakalanābhyām</i>      | “By addition and by subtraction”             |
| 8  | <i>Pūraṇāpūraṇābhyām</i>               | “By completing or non-completing”            |
| 9  | <i>Calana-Kalanābhyām</i>              | “By differentiation and integration”         |
| 10 | <i>Yāvadūnam</i>                       | “Whatever the extent of its deficiency”      |
| 11 | <i>Vyaṣṭi-Samaṣṭiḥ</i>                 | “Specific and general”                       |
| 12 | <i>Śeṣānyaṅkena Caramena</i>           | “The remainders by the last digit”           |
| 13 | <i>Sopāntya-dvayam Antyam</i>          | “The penultimate and the last”               |
| 14 | <i>Ekanyūnena Pūrveṇa</i>              | “By one less than the previous”              |
| 15 | <i>Guṇitasamuccayaḥ</i>                | “The product of the sum is the sum of the    |

| #  | Sanskrit                                           | English gloss         |
|----|----------------------------------------------------|-----------------------|
| 16 | <i>Guṇaka Samuccayaḥ</i>                           | products”             |
|    | Plus 13 sub-sūtras (including <i>Anurūpyeṇa</i> ). | “All the multipliers” |

## 2. Who was Tirthaji?

- **Bharati Krishna Tirthaji** (1884–1960). Śaṅkarācārya of Govardhana Maṭha (Puri) — i.e. head of one of the four major Hindu monastic seats founded by Ādi Śaṅkara.
- Studied at Madras Christian College. Multi-lingual scholar.
- Wrote the manuscript of *Vedic Mathematics* sometime between roughly 1911 and 1918 according to its editor’s foreword.
- The manuscript was lost or unfinished for decades.
- Tirthaji **died in 1960**.
- The book was **published in 1965**, five years after his death, edited by Vasudev Sharan Agrawala.

## 3. The provenance question. “Tirthaji claimed the 16 sūtras come from a Vedic text called the Atharva-veda Paṛiśiṣṭa — a supplement to the Atharva Veda.”

- **The problem:** No other Sanskritist has produced a manuscript of this *Paṛiśiṣṭa* containing these 16 sūtras.
- **The mathematician S. G. Dani** (TIFR, IIT Bombay) wrote a careful critique in 1993, reprinted in *Resonance* (October 2001), pointing out:
  - No manuscript evidence.
  - The Sanskrit style of the sūtras is more recent than Vedic.
  - The mathematical techniques have 19th–20th century parallels.

## 4. Honest framing. Two things can be true at once:

1. The  $\times 11$  trick (and the rest of Tirthaji’s 16-sūtra system) **works**. It is a useful, beautiful piece of mental arithmetic.
2. The trick is **not verifiably ancient Vedic mathematics**. The best current evidence is that Tirthaji synthesised it in the early 20th century, drawing on classical Indian arithmetic + his own pedagogical genius, and named the sūtras in Sanskrit.

The synthesis itself is a real Indian intellectual achievement. The “Vedic” naming is a 20th-century interpretive choice that scholars dispute.

## 5. What’s at stake. If the sūtras are Vedic (3000+ years old), they’re a major historical claim. If they’re 20th-century synthesis, **they’re still a real achievement** — just a different one. Tirthaji’s pedagogical genius is not diminished by being recent.

## Discussion (15 min) — Take a position

Pose two questions to the class. Take a brief show-of-hands; then take 4–5 student statements (no debate, just listening).

1. “Does it matter whether the trick is ‘ancient’ or ‘20th-century’? Why or why not?”
2. “If it turns out to be 20th-century, should we still teach it as ‘Vedic mathematics’? What should we call it?”

End by noting: “At Senior band you’ll read Dani’s critique in full and form a deeper position.”

## Wrap-up (2 min)

- Preview Day 9: “Tomorrow — mixed-practice day plus project work. Friday is the final assessment.”

## Student-facing content

### Tirthaji and the 16 sūtras

Bharati Krishna Tirthaji (1884–1960), Śaṅkarācārya of Govardhana Maṭha, wrote *Vedic Mathematics* between roughly 1911 and 1918. The book was published in 1965, five years after his death.

Tirthaji claimed the 16 sūtras (including our *Ekādhikena Pūrveṇa*) come from a Vedic appendix-text. **No other Sanskritist has produced this manuscript.** S. G. Dani (1993) argues the sūtras are best understood as a 20th-century synthesis with Indian-mathematics roots, not a verifiably ancient Vedic teaching.

The trick works. The history is contested. Both can be true. The Indian intellectual achievement is real either way.

## Homework

- Read the 16-sūtra list. Pick ONE sūtra (other than the two we’ve studied) and find what mathematical operation it’s used for. (Sūtras 2, 3, 4, 10, 14 are well-documented in Tirthaji 1965.) Write 3 sentences.

## Differentiation

- **Tier up:** Read a one-paragraph excerpt from Dani (1993). What is Dani’s strongest critique? What does he NOT criticise?
- **Tier down:** Memorise the names of sūtras 1, 2, 3, and 14 (the four most commonly taught). Skip the historicity discussion details.

## Teacher notes

- Some students (and their families) believe firmly that the trick is “ancient Vedic mathematics.” Honour the belief; don’t dismiss it. Frame your position as “the academic consensus is more sceptical, and you’ll read the evidence at Senior band.”
- Avoid civilisational chest-thumping (“ancient Indians knew this before everyone”). The Indian achievement here is the synthesis and clarity of Tirthaji’s pedagogical work; that’s enough to celebrate without overreach.

- Avoid the opposite error too — dismissing the trick as “just Western math with Sanskrit labels.” The pedagogical insight (the named patterns, the rhythm) is genuinely Tirthaji’s.

### Citation(s) used in this lesson

- Tirthaji, B. K. (1965). *Vedic Mathematics*. Motilal Banarsidass. — Sūtra list and preface.
- Dani, S. G. (1993). “Myths and Reality: On Vedic Mathematics.” Reprinted in *Resonance* (October 2001).
- *Microsoft Word - sutras1.pdf* (SutraGanita corpus) — the 16+13 sūtra list.